
PRIOR AUTHORIZATION TRIAGE & POLICY COMPANION

Output Explainability Demo (mock mode)

LLM Mode : LIVE - Qwen via LM Studio

APPROVE - Covered dx + J5 state (TX) [LCD Path]

REQUEST

Procedure Code : 64483

Diagnosis : M54.16

State : TX

Patient Age : 55

Clinical Notes : Patient presents with lumbar radiculopathy confirmed on MRI. Conservative therapy was tried for 8 weeks without relief.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED

Jurisdiction : TX -> MATCHED

LCD : L39054 Epidural Injections for Pain Management

POLICY EVIDENCE

Source : LCD L39054

Section : Indications

Text : Epidural steroid injections (ESIs) are covered when the patient has a diagnosis of radiculopathy (ICD-10: M54.16, M54.17) confirmed by clinical examination and supported by imaging (MRI or CT). The pr

Similarity : 0.9000

Source : LCD L39054

Section : Limitations

Text : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain w

Similarity : 0.9000

Source : LCD L39054

Section : Documentation Requirements

Text : Documentation must support the reasonable and necessary nature of the epidural injection. Required elements: (1) radiculopathy symptoms confirmed on MRI or CT imaging, (2) failure of conservative ther

Similarity : 0.9000

POLICY CRITERIA

C1 [SQL] LCD-L39054-Cf0713c

Requirement : patient has a diagnosis of radiculopathy (ICD-10: M54

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : patient has a diagnosis of radiculopathy (ICD-10: M54; LCD L39054 -> ICD-10 covered: M54.16

Patient Ev. : Submitted ICD-10: M54.16

Explanation : The submitted diagnosis code(s) M54.16 are present in the LCD L39054 covered ICD-10 data. The diagnosis

requirement is satisfied.

C2 [AGENTIC_QWEN] LCD-L39054-C6ac622
Requirement : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters with...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis_confirmation: Evidence of a confirmed structural diagnosis of spinal pathology, such as imaging studies or clinical findings supporting a specific spinal condition; Required: clinical_indication: Documentation showing a clinical indication of spinal pathology, including physician notes or clinical assessments indicating a specific spinal condition
Patient Ev. : Patient presents with lumbar radiculopathy confirmed on MRI
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain without a confirmed structural diagnosis are not covered. Maximum of 3 injections per 6-month period unless clinical documentation supports additional treatment.
Policy: LCD L39054

Required Evidence:
* diagnosis_confirmation: Evidence of a confirmed structural diagnosis of spinal pathology, such as imaging studies or clinical findings supporting a specific spinal condition
* clinical_indication: Documentation showing a clinical indication of spinal pathology, including physician notes or clinical assessments indicating a specific spinal condition
* treatment_frequency_record: Clinical documentation supporting the number of injections administered within a 6-month period, including dates and rationale for any injections beyond the three-per-6-month limit

Patient Evidence:
* Patient presents with lumbar radiculopathy confirmed on MRI

Missing Evidence:
* No documentation of the number of injections administered within a 6-month period
* No documentation supporting the rationale for injections beyond the three-per-6-month limit
* No clinical documentation showing a specific structural diagnosis of spinal pathology beyond 'lumbar radiculopathy confirmed on MRI' (which is not explicitly stated as a structural diagnosis of spinal pathology)

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 22866ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 7288ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 3 missing evidence items in 7420ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 3 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C3 [AGENTIC_QWEN] LCD-L39054-Cea7ddc

Requirement : Documentation must support the reasonable and necessary nature of the epidural injection

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: diagnosis_support: Evidence that the patient's diagnosis (e.g., M54.16) is consistent with a condition that warrants epidural injection; Required: clinical_indication: Documentation of clinical signs or symptoms indicating the need for epidural injection, such as pain severity or functional limitation

Patient Ev. : Patient presents with lumbar radiculopathy confirmed on MRI.; Conservative therapy was tried for 8 weeks without relief.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Documentation must support the reasonable and necessary nature of the epidural injection

Policy: LCD L39054

Required Evidence:

* diagnosis_support: Evidence that the patient's diagnosis (e.g., M54.16) is consistent with a condition that warrants epidural injection

* clinical_indication: Documentation of clinical signs or symptoms indicating the need for epidural injection, such as pain severity or functional limitation

* treatment_plan: Evidence of a documented treatment plan that includes epidural injection as a reasonable and necessary intervention

* alternative_treatment: Documentation showing that other conservative treatments have been tried and failed, supporting the need for epidural injection

Patient Evidence:

* Patient presents with lumbar radiculopathy confirmed on MRI.

* Conservative therapy was tried for 8 weeks without relief.

Missing Evidence:

* Documentation showing that the patient's diagnosis (M54.16) is consistent with a condition that warrants epidural injection

* Documentation of clinical signs or symptoms indicating the need for epidural injection, such as pain severity or functional limitation

* Evidence of a documented treatment plan that includes epidural injection as a reasonable and necessary intervention

* Documentation showing that other conservative treatments have been tried and failed, supporting the need for epidural injection

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 22909ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 4 required evidence categories in 8342ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 4 missing evidence items in 8082ms.

[EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 4 required evidence categories are missing. Evidence is insufficient for a definitive conclusion.

[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.

[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 1
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 2
Evidence Fusion Result : UNKNOWN

FINAL DECISION

[INFO] Decision : REQUEST_MORE_INFORMATION

Reason : The request is missing required information: Clinical documentation required to verify ambiguous criteria. Additional information must be provided before a coverage determination can be made. [MISSING_REQUIRED_INFORMATION]

Reason Codes: MISSING_REQUIRED_INFORMATION

Decision Basis:

Additional information is required to determine coverage.

* LCD-L39054-Cf0713c: SATISFIED

* LCD-L39054-C6ac622: UNKNOWN

* LCD-L39054-Cea7ddc: UNKNOWN

Evidence Fusion: UNKNOWN

DecisionEngine: UNKNOWN -> REQUEST_MORE_INFORMATION

PEND - Explicitly non-covered diagnosis [LCD Path]

REQUEST

Procedure Code : 64483

Diagnosis : Z00.00

State : TX

Patient Age : 40

Clinical Notes : Routine general examination. No structural pathology identified.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED

Jurisdiction : TX -> MATCHED

LCD : L39054 Epidural Injections for Pain Management

POLICY EVIDENCE

Source : LCD L39054

Section : Indications

Text : Epidural steroid injections (ESIs) are covered when the patient has a diagnosis of radiculopathy (ICD-10: M54.16, M54.17) confirmed by clinical examination and supported by imaging (MRI or CT). The pr

Similarity : 0.9000

Source : LCD L39054

Section : Limitations

Text : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain w

Similarity : 0.9000

Source : LCD L39054

Section : Documentation Requirements

Text : Documentation must support the reasonable and necessary nature of the epidural injection. Required elements: (1)

radiculopathy symptoms confirmed on MRI or CT imaging, (2) failure of conservative ther
Similarity : 0.9000

POLICY CRITERIA

C1 [SQL] LCD-L39054-C3e4101
Requirement : patient has a diagnosis of radiculopathy (ICD-10: M54
Type : STRUCTURED
Result : NOT_SATISFIED
Policy Ev. : patient has a diagnosis of radiculopathy (ICD-10: M54; LCD L39054 -> ICD-10 non-covered: Z00.00
Patient Ev. : Submitted ICD-10: Z00.00
Explanation : The submitted diagnosis code(s) Z00.00 are explicitly listed in the LCD L39054 non-covered ICD-10 data. The diagnosis requirement is not satisfied.

C2 [AGENTIC_QWEN] LCD-L39054-C06ead1
Requirement : The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters witho...
Type : SEMANTIC
Result : NOT_SATISFIED
Policy Ev. : Required: diagnosis_specification: Evidence that a confirmed structural diagnosis of spinal pathology exists, such as radiographic or imaging findings indicating a specific spinal condition; Required: clinical_indication: Documentation showing a clinical indication of spinal pathology, including symptoms, physical exam findings, and clinical judgment supporting the need for injection therapy
Patient Ev. : Routine general examination. No structural pathology identified.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: The following diagnoses are non-covered under this LCD: routine general examination (Z00.00), encounters without clinical indication of spinal pathology. Injections performed solely for general pain without a confirmed structural diagnosis are not covered. Maximum of 3 injections per 6-month period unless clinical documentation supports additional treatment.
Policy: LCD L39054

Required Evidence:
* diagnosis_specification: Evidence that a confirmed structural diagnosis of spinal pathology exists, such as radiographic or imaging findings indicating a specific spinal condition
* clinical_indication: Documentation showing a clinical indication of spinal pathology, including symptoms, physical exam findings, and clinical judgment supporting the need for injection therapy
* treatment_frequency_record: Clinical documentation indicating the number of injections administered within the 6-month period and justification for any injections beyond the maximum of three

Patient Evidence:
* Routine general examination. No structural pathology identified.

Missing Evidence:
* Confirmed structural diagnosis of spinal pathology (e.g., radiographic or imaging findings indicating a specific spinal condition)
* Clinical indication of spinal pathology, including symptoms, physical exam findings, and clinical judgment supporting the need for injection therapy
* Documentation of the number of injections administered within the 6-month period and justification for any injections beyond the maximum of three

Qwen Result: NOT_SATISFIED
Critic Result: VALIDATED
Final Result: NOT_SATISFIED

The submitted clinical documentation does not satisfy this semantic policy criterion based on available evidence.

(Agentic pipeline completed in 19800ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 8694ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 3 missing evidence items in 5711ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 3 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned NOT_SATISFIED. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: NOT_SATISFIED. All 6 checks passed.

C3 [AGENTIC_QWEN] LCD-L39054-Ca4ba57
Requirement : Documentation must support the reasonable and necessary nature of the epidural injection
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: diagnosis_support: Evidence that the patient's diagnosis (e.g., Z00.00) is consistent with a condition that warrants epidural injection; Required: clinical_indication: Documentation of clinical signs or symptoms indicating the need for epidural injection
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: Documentation must support the reasonable and necessary nature of the epidural injection
Policy: LCD L39054

Required Evidence:

- * diagnosis_support: Evidence that the patient's diagnosis (e.g., Z00.00) is consistent with a condition that warrants epidural injection
- * clinical_indication: Documentation of clinical signs or symptoms indicating the need for epidural injection
- * treatment_plan: Evidence that the epidural injection is part of a reasonable and necessary treatment plan
- * medical_necessity: Documentation demonstrating that the procedure is medically necessary based on clinical judgment

Patient Evidence:

- * (No supporting evidence found in request)

Missing Evidence:

- * No documentation of diagnosis support for Z00.00 being consistent with a condition warranting epidural injection
- * No documentation of clinical signs or symptoms indicating the need for epidural injection
- * No documentation that the epidural injection is part of a reasonable and necessary treatment plan
- * No documentation demonstrating medical necessity of the procedure based on clinical judgment

Qwen Result: NOT_SATISFIED
Critic Result: REJECTED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 17460ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 4 required evidence categories in 7685ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 0 supporting, 0 contradicting, 4 missing evidence items in 6159ms.

[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 4 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned NOT_SATISFIED. Evidence cited: 0 items.
[CRITIC_AGENT] COMPLETED: Critic REJECTED Qwen's result (NOT_SATISFIED). Final result: UNKNOWN. Reasons: Qwen returned NOT_SATISFIED when no contradicting evidence is present - only missing evidence. The correct result should be UNKNOWN, not NOT_SATISFIED (absence of evidence evidence of absence).

EVIDENCE FUSION

Criteria SATISFIED : 0
Criteria NOT_SATISFIED : 2
Criteria UNKNOWN : 1
Evidence Fusion Result : EXCLUDED

FINAL DECISION

[PEND] Decision : PEND
Reason : The submitted procedure is explicitly excluded by the applicable Local Coverage Determination (LCD). The request is pending for manual review. [LCD_EXCLUDES_PROCEDURE]
Reason Codes: LCD_EXCLUDES_PROCEDURE

Decision Basis:
The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.
* LCD-L39054-C3e4101: NOT_SATISFIED
* LCD-L39054-C06ead1: NOT_SATISFIED
* LCD-L39054-Ca4ba57: UNKNOWN
Evidence Fusion: EXCLUDED
DecisionEngine: EXCLUDED -> PEND

APPROVE - Stem Cell Transplant covered by NCD 110.23 [NCD RAG Path]

REQUEST

Procedure Code : 38240
Diagnosis : C91.10
State : TX
Patient Age : 52
Clinical Notes : Patient diagnosed with chronic lymphocytic leukemia (CLL), relapsed after first-line therapy. Allogeneic hematopoietic stem cell transplantation recommended as curative intent option. Patient has good performance status and adequate cardiopulmonary function documented.

POLICY IDENTIFICATION

NCD : NCD-110.23
Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD NCD-110.23
Section : Coverage Indications
Text : Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, Aplastic Anemia, Severe Comb

Similarity : 0.9000
Source : NCD NCD-110.23
Section : Coverage with Evidence Development
Text : Allo-HSCT is covered for Medicare beneficiaries with the following indications ONLY when participating in approved prospective clinical studies meeting CED criteria: Myelodysplastic Syndromes, Multipl
Similarity : 0.9000
Source : NCD NCD-110.23
Section : Documentation Requirements
Text : Documentation to support the reasonable and necessary nature of allo-HSCTs must speak to the serious illness of the patient, reasons why the patient is considered to have relapsed or refractory diseases
Similarity : 0.9000

CODE MATCHING

HCPCS 38240 -> NCD NCD-110.23 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-NCD-110.23-HCPCS
Requirement : The requested procedure must be an applicable service under NCD NCD-110.23.
Type : STRUCTURED
Result : SATISFIED
Policy Ev. : NCD NCD-110.23 contains HCPCS 38240 in its covered-procedure list.
Patient Ev. : Submitted HCPCS: 38240
Explanation : Procedure 38240 is listed in NCD NCD-110.23 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-NCD-110.23-C229470
Requirement : is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, A...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, Aplastic Anemia, Severe Combined Immunodeficiency disease (SCID), and Wiskott-Aldrich Syndrome
Patient Ev. : Patient diagnosed with chronic lymphocytic leukemia (CLL), relapsed after first-line therapy.; Allogeneic hematopoietic stem cell transplantation recommended as curative intent option.; Patient has good performance status and adequate cardiopulmonary function documented.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for Medicare beneficiaries with the following conditions when reasonable and necessary: Leukemia, Aplastic Anemia, Severe Combined Immunodeficiency disease (SCID), and Wiskott-Aldrich Syndrome
Policy: NCD NCD-110.23

Required Evidence:
* (Policy Agent could not identify specific categories)

Patient Evidence:
* Patient diagnosed with chronic lymphocytic leukemia (CLL), relapsed after first-line therapy.
* Allogeneic hematopoietic stem cell transplantation recommended as curative intent option.
* Patient has good performance status and adequate cardiopulmonary function documented.

Missing Evidence:
* Specific diagnosis of leukemia, aplastic anemia, severe combined immunodeficiency disease (SCID), or Wiskott-Aldrich syndrome

- * Evidence that the treatment is reasonable and necessary
- * Evidence of the specific indication for allogeneic hematopoietic stem cell transplantation beyond being recommended as curative intent
- * Documentation of the patient's eligibility under the policy based on diagnosis and clinical context

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 19238ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 0 required evidence categories in 3842ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 3 supporting, 0 contradicting, 4 missing evidence items in 8339ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 3 supporting items found, but 4 required evidence categories are missing. Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 3 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C3 [AGENTIC_QWEN] NCD-NCD-110.23-C6762f0
Requirement : is covered for Medicare beneficiaries with the following indications ONLY when participating in approved prosp...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: clinical_study_approval: Documentation that the clinical study is approved and meets CED (Clinical Evidence Development) criteria; Required: diagnosis_specificity: Evidence confirming the patient's diagnosis matches one of the specified conditions: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or high risk, or severe and symptomatic sickle cell disease
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: is covered for Medicare beneficiaries with the following indications ONLY when participating in approved prospective clinical studies meeting CED criteria: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or High risk, Sickle cell disease that is severe and symptomatic
Policy: NCD NCD-110.23

Required Evidence:
* clinical_study_approval: Documentation that the clinical study is approved and meets CED (Clinical Evidence Development) criteria
* diagnosis_specificity: Evidence confirming the patient's diagnosis matches one of the specified conditions: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or high risk, or severe and symptomatic sickle cell disease
* prospective_study_participation: Proof that the patient is actively participating in an approved prospective clinical study

Patient Evidence:
* (No supporting evidence found in request)

Missing Evidence:
* No documentation that the clinical study is approved and meets CED (Clinical Evidence Development) criteria
* No evidence confirming the patient's diagnosis matches one of the specified conditions: Myelodysplastic Syndromes, Multiple myeloma (Durie-Salmon Stage II or III), Myelofibrosis with DIPSS-plus intermediate-2 or high risk, or severe and symptomatic sickle cell disease; patient is diagnosed with chronic lymphocytic leukemia (CLL), which is not listed

* No proof that the patient is actively participating in an approved prospective clinical study

Qwen Result: NOT_SATISFIED

Critic Result: REJECTED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 21406ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 8796ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 0 supporting, 0 contradicting, 3 missing evidence items in 7262ms.

[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 3 required evidence categories are missing.

[QWEN] COMPLETED: Qwen returned NOT_SATISFIED. Evidence cited: 0 items.

[CRITIC_AGENT] COMPLETED: Critic REJECTED Qwen's result (NOT_SATISFIED). Final result: UNKNOWN. Reasons: Qwen returned NOT_SATISFIED when no contradicting evidence is present - only missing evidence. The correct result should be UNKNOWN, not NOT_SATISFIED (absence of evidence evidence of absence).

C4 [RULES] NCD-NCD-110.23-C6bfa6a

Requirement : Documentation to support the reasonable and necessary nature of allo-HSCTs must speak to the serious illness o...

Type : RULE_BASED

Result : UNKNOWN

Policy Ev. : Documentation to support the reasonable and necessary nature of allo-HSCTs must speak to the serious illness of the patient, reasons why the patient is considered to have relapsed or refractory disease, relevant clinical contextual information such as age, frailty, performance status, cardiopulmonary function, and psychosocial support structures.

Patient Ev. : No explicit deterministic rule logic implemented for this text.

Explanation : A rule-based requirement was identified but no matching evaluation logic exists for this criterion text. The requirement cannot be evaluated.

EVIDENCE FUSION

Criteria SATISFIED : 1

Criteria NOT_SATISFIED : 0

Criteria UNKNOWN : 3

Evidence Fusion Result : UNKNOWN

FINAL DECISION

[PEND] Decision : PEND

Reason : Policy evidence was ambiguous or conflicting. Manual clinical review is required to determine coverage.

[AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW]

Reason Codes: AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW

Decision Basis:

The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.

* NCD-NCD-110.23-HCPCS: SATISFIED

* NCD-NCD-110.23-C229470: UNKNOWN

* NCD-NCD-110.23-C6762f0: UNKNOWN

* NCD-NCD-110.23-C6bfa6a: UNKNOWN

Evidence Fusion: UNKNOWN

DecisionEngine: UNKNOWN -> PEND

APPROVE - AFP lab test covered by NCD 190.25 [NCD RAG Path]

REQUEST

Procedure Code : 82105

Diagnosis : C22.0

State : CA

Patient Age : 67

Clinical Notes : Hepatocellular carcinoma in high-risk patient with alcoholic cirrhosis. AFP serum test ordered to monitor response to treatment.

POLICY IDENTIFICATION

NCD : NCD-190.25

Jurisdiction : CA -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD NCD-190.25

Section : Indications

Text : AFP (Alpha-fetoprotein) testing (HCPCS 82105, 82106) is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic cirrhosis, cirrhosis of viral etiology,

Similarity : 0.9000

CODE MATCHING

HCPCS 82105 -> NCD NCD-190.25 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-NCD-190.25-HCPCS

Requirement : The requested procedure must be an applicable service under NCD NCD-190.25.

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : NCD NCD-190.25 contains HCPCS 82105 in its covered-procedure list.

Patient Ev. : Submitted HCPCS: 82105

Explanation : Procedure 82105 is listed in NCD NCD-190.25 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-NCD-190.25-C4be5a9

Requirement : is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic ci...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic cirrhosis, cirrhosis of viral etiology, hemochromatosis, and alpha 1-antitrypsin deficiency

Patient Ev. : Hepatocellular carcinoma in high-risk patient with alcoholic cirrhosis

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for the diagnosis of hepatocellular carcinoma in high-risk patients such as those with alcoholic cirrhosis,

cirrhosis of viral etiology, hemochromatosis, and alpha 1-antitrypsin deficiency
Policy: NCD NCD-190.25

Required Evidence:
* (Policy Agent could not identify specific categories)

Patient Evidence:
* Hepatocellular carcinoma in high-risk patient with alcoholic cirrhosis

Missing Evidence:
* Evidence of viral etiology cirrhosis
* Evidence of hemochromatosis
* Evidence of alpha 1-antitrypsin deficiency
* Evidence of AFP serum test result or monitoring data
* Evidence of diagnosis confirmation beyond C22.0

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 13450ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 0 required evidence categories in 3581ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 5 missing evidence items in 5401ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 5 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 1
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 1
Evidence Fusion Result : UNKNOWN

FINAL DECISION

[PEND] Decision : PEND
Reason : Policy evidence was ambiguous or conflicting. Manual clinical review is required to determine coverage.
[AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW]
Reason Codes: AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW

Decision Basis:
The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.
* NCD-NCD-190.25-HCPCS: SATISFIED
* NCD-NCD-190.25-C4be5a9: UNKNOWN
Evidence Fusion: UNKNOWN
DecisionEngine: UNKNOWN -> PEND

APPROVE - TENS covered by NCD 160.7.1 [NCD RAG + NCD Decision Path]

REQUEST

Procedure Code : 64550

Diagnosis : G89.29

State : TX

Patient Age : 48

Clinical Notes : Patient presents with chronic pain syndrome following lumbar surgery. Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief. TENS (transcutaneous electrical nerve stimulation) requested as adjunct pain management.

POLICY IDENTIFICATION

NCD : N123

Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD N123

Section : Coverage Indications

Text : Transcutaneous Electrical Nerve Stimulation (TENS) (HCPCS 64550) is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried and failed. Documentatio

Similarity : 0.9000

Source : NCD N123

Section : Limitations

Text : TENS is not covered as initial therapy for acute pain; conservative measures must be attempted first. TENS for chronic non-cancer pain requires documented failure of conventional pharmacological treat

Similarity : 0.9000

CODE MATCHING

HCPCS 64550 -> NCD N123 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-N123-HCPCS

Requirement : The requested procedure must be an applicable service under NCD N123.

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : NCD N123 contains HCPCS 64550 in its covered-procedure list.

Patient Ev. : Submitted HCPCS: 64550

Explanation : Procedure 64550 is listed in NCD N123 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-N123-C147a00

Requirement : Documentation must support that the patient has a diagnosis of chronic pain requiring neurostimulation therapy...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: diagnosis: Evidence that the patient has a diagnosis of chronic pain, specifically coded as G89.29; Required:

treatment_failure: Documentation showing that conventional pharmacological treatments have been inadequate or contraindicated

Patient Ev. : Patient presents with chronic pain syndrome following lumbar surgery.; Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Documentation must support that the patient has a diagnosis of chronic pain requiring neurostimulation therapy and that conventional pharmacological treatment has been inadequate or contraindicated

Policy: NCD N123

Required Evidence:

- * diagnosis: Evidence that the patient has a diagnosis of chronic pain, specifically coded as G89.29
- * treatment_failure: Documentation showing that conventional pharmacological treatments have been inadequate or contraindicated

Patient Evidence:

- * Patient presents with chronic pain syndrome following lumbar surgery.
- * Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Missing Evidence:

- * No explicit documentation that conventional pharmacological treatments have been inadequate or contraindicated beyond 'without satisfactory relief'. The policy requires documentation of failure or contraindication, which is not explicitly stated.
- * No explicit statement confirming the diagnosis is coded as G89.29, though the diagnosis is listed in additional structured facts. The patient data does not contain a direct statement of the diagnosis coded as G89.29.

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 19456ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 5553ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 2 missing evidence items in 7091ms.

[EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 2 required evidence categories are missing.

Evidence is insufficient for a definitive conclusion.

[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.

[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C3 [AGENTIC_QWEN] NCD-N123-C17bb32

Requirement : patient has a diagnosis of chronic pain requiring neurostimulation therapy and that conventional pharmacologic...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: diagnosis: Evidence of a chronic pain diagnosis, specifically coded as G89.29 or equivalent; Required:

treatment_response: Documentation that conventional pharmacological treatments have been inadequate or contraindicated

Patient Ev. : Patient presents with chronic pain syndrome following lumbar surgery.; Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: patient has a diagnosis of chronic pain requiring neurostimulation therapy and that conventional pharmacological

treatment has been inadequate or contraindicated

Policy: NCD N123

Required Evidence:

- * diagnosis: Evidence of a chronic pain diagnosis, specifically coded as G89.29 or equivalent
- * treatment_response: Documentation that conventional pharmacological treatments have been inadequate or contraindicated
- * therapy_indication: Clinical indication that neurostimulation therapy is necessary for the patient's chronic pain

Patient Evidence:

- * Patient presents with chronic pain syndrome following lumbar surgery.
- * Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Missing Evidence:

- * Documentation that conventional pharmacological treatments have been inadequate or contraindicated (specifically stating failure or unsuitability of standard pain medications)
- * Clinical indication that neurostimulation therapy is necessary for the patient's chronic pain (only TENS is mentioned as requested, not neurostimulation therapy)
- * Required evidence not found in request: 'Diagnoses: G89.29'

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 20278ms)

Agent Trace:

- [POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 6715ms.
- [CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 3 missing evidence items in 7690ms.
- Removed 1 unsupported inferences.
- [EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 3 required evidence categories are missing.
- Evidence is insufficient for a definitive conclusion.
- [QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.
- [CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C4 [AGENTIC_QWEN] NCD-N123-Caf558c

Requirement : is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried and failed

Patient Ev. : Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for acute post-operative pain and chronic intractable pain when conservative therapy has been tried and failed

Policy: NCD N123

Required Evidence:

- * (Policy Agent could not identify specific categories)

Patient Evidence:

- * Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Missing Evidence:

- * Evidence that the pain is acute post-operative pain or chronic intractable pain
- * Evidence that conservative treatments other than pharmacological therapy have been tried and failed
- * Evidence that TENS is being used as a treatment for pain management

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 12750ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 0 required evidence categories in 3033ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 3 missing evidence items in 5226ms.

[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 3 required evidence categories are missing.

Evidence is insufficient for a definitive conclusion.

[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.

[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C5 [AGENTIC_QWEN] NCD-N123-C4fea46

Requirement : must have failed conservative therapy for >= 6 weeks

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: conservative_therapy_documentation: Evidence that the patient received and completed conservative therapy (e.g., physical therapy, medication, lifestyle changes) as prescribed, with documented dates and duration of therapy;

Required: therapy_failure_confirmation: Clinical documentation confirming that the conservative therapy failed to improve symptoms or outcomes, including physician assessments or objective clinical findings

Patient Ev. : Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: must have failed conservative therapy for >= 6 weeks

Policy: NCD N123

Required Evidence:

- * conservative_therapy_documentation: Evidence that the patient received and completed conservative therapy (e.g., physical therapy, medication, lifestyle changes) as prescribed, with documented dates and duration of therapy
- * therapy_failure_confirmation: Clinical documentation confirming that the conservative therapy failed to improve symptoms or outcomes, including physician assessments or objective clinical findings
- * duration_of_failure: Documentation showing that the failure of conservative therapy lasted for at least 6 weeks, including start and end dates of the failure period

Patient Evidence:

- * Conservative pharmacological therapy has been tried for over 6 weeks without satisfactory relief.

Missing Evidence:

- * Documentation showing the specific dates and duration of conservative therapy (e.g., physical therapy, medication, lifestyle

changes) as prescribed

- * Clinical documentation confirming that conservative therapy failed to improve symptoms or outcomes, including physician assessments or objective clinical findings
- * Documentation showing the start and end dates of the failure period of conservative therapy, demonstrating a duration of at least 6 weeks

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 22199ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 7489ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 3 missing evidence items in 8280ms.

[EVALUATION_AGENT] COMPLETED: Partial evidence: 1 supporting items found, but 3 required evidence categories are missing. Evidence is insufficient for a definitive conclusion.

[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.

[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 1
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 4
Evidence Fusion Result : UNKNOWN

FINAL DECISION

[PEND] Decision : PEND
Reason : Policy evidence was ambiguous or conflicting. Manual clinical review is required to determine coverage.
[AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW]
Reason Codes: AMBIGUOUS_EVIDENCE_REQUIRES_REVIEW

Decision Basis:

The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.

- * NCD-N123-HCPCS: SATISFIED
- * NCD-N123-C147a00: UNKNOWN
- * NCD-N123-C17bb32: UNKNOWN
- * NCD-N123-Caf558c: UNKNOWN
- * NCD-N123-C4fea46: UNKNOWN

Evidence Fusion: UNKNOWN
DecisionEngine: UNKNOWN -> PEND

REQUEST_MORE_INFORMATION - Outside jurisdiction (CA) [LCD Path]

REQUEST

Procedure Code : 64483
Diagnosis : M54.17
State : CA
Patient Age : 63
Clinical Notes : Lumbosacral radiculopathy.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED
Jurisdiction : CA -> NOT_MATCHED

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

EVIDENCE FUSION

Criteria SATISFIED : 0
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 0
Evidence Fusion Result : NOT_ADDRESSED

FINAL DECISION

[INFO] Decision : REQUEST_MORE_INFORMATION
Reason : The request is missing required information: State outside jurisdiction. Additional information must be provided before a coverage determination can be made. [MISSING_REQUIRED_INFORMATION]
Reason Codes: MISSING_REQUIRED_INFORMATION

Decision Basis:
Additional information is required to determine coverage.
* No formal criteria were evaluated (deterministic code matching applied).
Evidence Fusion: NOT_ADDRESSED
DecisionEngine: NOT_ADDRESSED -> REQUEST_MORE_INFORMATION

REQUEST_MORE_INFORMATION - No matching policy

REQUEST

Procedure Code : 99999
Diagnosis : M54.16
State : TX
Patient Age : 45
Clinical Notes : Unknown procedure code.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED
Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

EVIDENCE FUSION

Criteria SATISFIED : 0

Criteria NOT_SATISFIED : 0

Criteria UNKNOWN : 0

Evidence Fusion Result : NOT_ADDRESSED

FINAL DECISION

[INFO] Decision : REQUEST_MORE_INFORMATION

Reason : No coverage policy was found that references procedure code '99999'. Prior authorization cannot be evaluated without an applicable policy.

Reason Codes: POLICY_NOT_FOUND

Decision Basis:

No applicable policy was found for procedure '99999'. Evidence Fusion: NOT_ADDRESSED. DecisionEngine: NOT_ADDRESSED
-> REQUEST_MORE_INFORMATION.

END OF DEMO
